KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 2nd Term Examination, 2020
ESE 3203
(Petroleum and Natural Gas Processing)

Time: 1 Hour 30 Minutes | Full Marks: 120

N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Classify and briefly describe the basic types of processes and operations in petroleum refining. 10

1(b). Explain the following terms: 10
i) BPSD, ii) BPCD, iii) Refinery Complexity ,
iv) °API, v) K_w

1(c). What is natural gas condensate? With the help of a flow diagram, show the separation of condensate from raw natural gas. 10

2(a). Classify different grades of gasoline based on octane number. 06

2(b). What are the components of the gasoline pool? Why are oxygenates added to gasoline? Name a few such 0xygenates. 10

2(c). Explain vacuum distillation process with the help of a flow diagram. 14

3(a). What is catalyst poisoning? Give examples of various types of catalyst poisons? 10

3(b). With a neat sketch, explain the hydrotreating process. 10

3(c). Why is hydrogen required in a refinery? From which sources can this hydrogen be sourced? 10

SECTION – B

4(a).	Explain Natural gas hydrates with diagram.	15
4(b).	Write a short note on "LPG and LNG"	05
4(c).	Write down the reasons for removal of H_2S and CO_2 in natural gas processing.	10
5(a).	"Natural gas is a clean burning fuel." –Explain it.	06
5(b).	Write a comparative analysis among saturated black oil reservoirs, under saturated black oil reservoirs and volatile oil reservoirs. Give appropriate figures if necessary.	18
5(c).	Why is dehydration process a must in natural gas processing?	06
6(a).	Briefly describe a conventional gas-liquid separator with schematic diagram.	10
6(b).	Find out pseudo-reduced pressure and pseudo-reduced temperature for the following gas composition:	20

$C_1 = 90\%,\quad C_2 = 3\%,\quad C_3 = 2\%,\quad H_2S = 2\%,\quad N_2 = 3\%$

The temperature & pressure of the gas are $200°F$ and 2000 psia respectively.

Also find out the density and formation volume factor of the gas, assuming compressibility factor of the gas is 0.92.

For necessary data, use the following table.

Table 6(b): Physical constants for typical natural gas constituents.

Compound	Molecular Weight	Critical Pressure(psia)	Critical Temperature (°R)
CH_4	16.043	667.8	343.1
C_2H_6	30.070	707.8	549.8
C_3H_8	44.097	616.3	665.7
n-C_4H_{10}	58.124	550.7	765.4
i-C_4H_{10}	58.124	529.1	734.7
CO_2	44.010	1070.9	547.6
H_2S	34.076	1306.0	672.4
N_2	28.013	493.0	227.3

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 2nd Term Examination, 2020
ESE 3221
(Energy Storage Systems)

Time: 1 hour 30 min. Full Marks: 120

N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

Section A

1(a). List the different energy storage systems. Distinguish between energy density and power density. 10

1(b). Describe the small-scale compressed air energy storage system. What are the advantages of flywheel energy storage system? 12

1(c). Draw a thematic of flywheel energy storage system and show its componts. 08

2(a). Classify thermal energy storage syetem. Describe the thermal energy storage system in solid storage medium. 10

2(b). What is hybrid power plants meant? Draw a thematic of hybrid power plant. 10

2(c). What are the characteristics of phase change materials? What are the advantages and disadvanctages of organic phase change materials? 10

3(a). What is a Super-Conducting Magnet Energy Storage System (SMES)? What are the advantages and disadvantages of SMES system? 10

3(b). What is a super capacitor? What are advantages of a super capacitor? 08

3(c). Compare capacitor with battery. 04

3(d). Write short notes on: 08
(i). Fast charging stations (ii). Online UPS

Section B

4(a).	Describe the classification of battery. What are the advantages of lead-acid battery?	11
4(b).	Explain the working principle of a Li-ion battery. How to recycle Li-ion batteries?	11
4(c).	What happen to anode and cathod of lithium ion battery in ageing?	08
5(a).	How does Vanadium redox flow battery (VRFB) work? Explain with reaction mechanisms.	11
5(b).	"VRFB is grid fiendly" – explain.	08
5(c).	Write down the applications and common challenges associated with nickel metal hydride battery.	11
6(a).	What are the differences between fuel cell and battery?	08
6(b).	Explain the working principle of a PEM fuel cell. What are the benefits of a PEM fuel cell?	12
6(c).	"Using metal hydride is one of the attactive and convenient way to store energy in a small fuel cell" – explain.	10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 2nd Term Examination, 2020
ESE 3207
(Solar Photovoltaic Systems)

Time: 1 Hour 30 Minutes — Full Marks: 120

N.B. i) Answer any two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). A solar cell with dimensions 12cm x 12cm is illuminated at standard test conditions. The cell has the following external parameters: 06

$V_{OC} = 0.8\,V, I_{SC} = 3\,A, V_{MPP} = 0.75\,V,\ I_{MPP} = 2.5\,A$. Calculate the fill factor and efficiency of this solar cell.

1(b). Explain the basic two transport mechanisms in a semiconductor device. 12

1(c). With the help of mathematical equations, explain the influence of series and shunt resistance on the performance of solar cell. 12

2(a). Define doping. Discuss the p-n junctions working procedure of solar cell. 11

2(b). How the tranparency loss and excess excitation loss can be minimized in solar cell? Explain in details. What are the main challenges in multi junction solar cell? Explain. 15

2(c). Write the diode equation and draw I-V curve of a ideal diode. 04

3(a). Explain the working principle of high efficiency Perovskite solar cells. 14

3(b). Consider a Si solar cell operating at 300K with the following parameters: 10

Area A = 1.5 cm^2, Accepton doping $N_a = 5 \times 10^{17}\text{cm}^{-3}$, Donar doping $= 8 \times 10^{17}\text{cm}^{-3}$, Electron diffusion coefficient $D_n = 22\ \text{cm}^2/\text{s}$, Hole diffusion coefficient $D_p = 10\text{cm}^2/\text{s}$, Electron recombination time $\tau_n = 3 \times 10^{-7}\text{s}$, Hole recombination time $\tau_p = 5 \times 10^{-7}\text{s}$, Electron hole pair generation rate by light $a_L = 10^{22}\text{cm}^{-3}\text{s}^{-1}$. Calculate the photo current. The solar cell is reversed biased with a voltage of 2V.

3(c). Why supporting layers are used in solar cells? Discuss. 06

SECTION – B

4(a). What is meant by Lithography? Write down the steps of Lithography with proper diagram. 12

4(b). Describe in details to make polished wafer of Si from the starting materials. 18

5(a). "Performance of solar cell depends on irradiance". Justify the statement. 08

5(b). Explain shedding of PV system. With proper justification, explain the performance of PV system during shedding condition. 16

5(c). Write down the component of on-grid PV system. 06

6(a). Write down the design guidelines of all components of a 3 kW off grid PV system. 12

6(b). Why Energy Payback time is important of a PV system design. 05

6(c). Compare the life cycle cost of a highway construction warning sign that is PV powered vs using a gasoline generator to power the same sign. 13

The system is capable of 24 hours per day operation with minimal down time. Assume the load to be 3 kWh per day with a 20 years life time. To power this load with a PV system, it will take 750-watt array of PV modules at a cost of $ 4 per watt, $900 worth of storage batteries, which need to be replaced energy 5 years, and a $ 300 charge controller. Assume a system maintenance cost of $100 per year. Assume a 500-watt gasoline generator can be purchased for $250. It needs about 545 gallons of gasoline per year with an annual maintenance cost of about $1500. Because of the heave use after energy 5 years, the generator must be replaced. Assume an in inflation rate of 3% and a discount rate of 10%.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 2nd Term Examination, 2020
ESE 3217
(Instrumentation and Control)

Time: 1 Hour 30 Minutes | Full Marks: 120

N.B. i) Answer any TWO questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

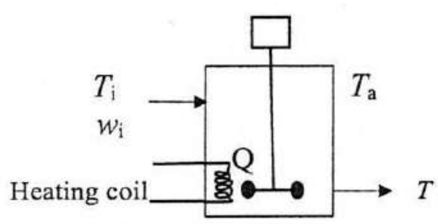
1(a) Discuss in brief the working principle of following instruments: 10
   i. Diaphragm pressure gauge.
   ii. Venturi tube

1(b) What are the major components of a control valve? Describe the working principle of quick-opening and equal-percentage valves with flow vs valve steam lift graph. 12

1(c) Define the following terms: 08
   i. Controlled variable
   ii. Disturbance variable

2(a) With block diagram describe working principle of a feedback control system. 10

2(b) A completely enclosed stirred-tank heating process is used to heat an incoming stream whose flow rate varies. The heating rate from this coil and volume are both constant. 20



ρ and C_p are constant
U overall heat transfer coefficient constant
A_s is the surface area for heat loss to ambient
$T_i > T_a$

(i) Develop a mathematical model that describes the exit temperature if heat loss to ambient occur and if the ambient temperature (T_a) and incoming stream's temperature (T_i) both can vary.

(ii) Discuss qualitatively what you expect to happen as T_i and w increases (or decreases).

3(a) Discuss the hierarchy of process control activities. 10

3(b) What is constant function and step function? Solve the differential equation $5\frac{dy}{dt} + 4y = 2,\ y(0) = 1$ using Laplace transforms. 10

3(c) Write short note on Additive property and Multiplicative properly of transfer functions. 10

SECTION – B

4(a) Explain the operations on Fuzzy sets. 12

4(b) Write the difference between PD and PID controller. 08

4(c) Discuss the different types of memory of a microcontroller. 10

5(a) What are the functions of SCAD system? Write the working procedure of SCAD system 11

5(b) Write the characteristics of Analog and Digital Signals. 08

5(c) Write a PLC program, a motor is switched on by pressing a spring-return push button start switch and the motor remains on until another spring return push button stop switch is pressed. 11

6(a) What is Master control relay of a PLC? Write a PLC program use three sensors and control a tire alarm system. 12

6(b) Write the scanning procedure of the ladder program. 08

6(c) What are the input and outputs of a PLC? 'A PLC can handle a large voltage spike'- Do you agree? Justify your answer. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 3rd Year 2nd Term Examination, 2020
ESE 3211
(Coal Power Generation)

Time: 1 Hour 30 Minutes. Full Marks: 120

N.B. i) Answer any Two questions from each section.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Write a short note on overview of coal power in Bangladesh. 10

1(b). What is rank of coal? Classify coal according to rank. 08

1(c). What are meant by followings: 12

(i) Coal handling (ii) Ash handling (iii) Coal Preparation (iv) Coal storage

2(a). With a schematic diagram, describe the working principle of the underground mining used when the coal seam outcrops at the surface. 15

2(b). Define bottom ash and fly ash. Write down the advantages and disadvantages of the ash found as a byproduct of burning of pulverized coal. 10

2(c). List the different types of dust collector systems. 05

3(a). How does circulating fluidized bed combustion technology reduce the emission of pollutants? Discuss. 15

3(b). Why secondary air is supplied in coal combustion? 05

3(c). Make a comparison between unit and central system of burning of pulverized coal. 10

SECTION – B

4(a). What is a steam generator? If a steam generator is needed to purchase, what are the different information should be considered? 08

4(b). Why mountings are needed in a boiler for its safe operation? 04

4(c). A boiler plant consumes 4450 kg of fuel per hour. The boiler evaporates 37000 kg of water per hour at a pressure of 21 bar and temperature of 265 °C. Temperature of water entering the economizer is 20 °C and temperature of water leaving the economizer is 95 °C. If the calorific value of fuel is 29500 kJ/kg, determine: 18

(i) The equivalent evaporation rate per kg of fuel
(ii) The thermal efficiency of plant
(iii) The percentage of heat utilized in boiler and economizer respectively.

Following data can be used:

Enthalpy of water at 20 °C = 83.914 kJ/kg
Enthalpy of water at 95 °C = 398.105 kJ/kg
Enthalpy of steam at 21 bar = 2798.18 kJ/kg
Saturation temperature of steam at 21 bar = 214.85 °C

5(a). Why does pressure in a nozzle decreases as the fluid velocity increases? 06

5(b). In contrast of steam turbine, what are meant by followings: 09

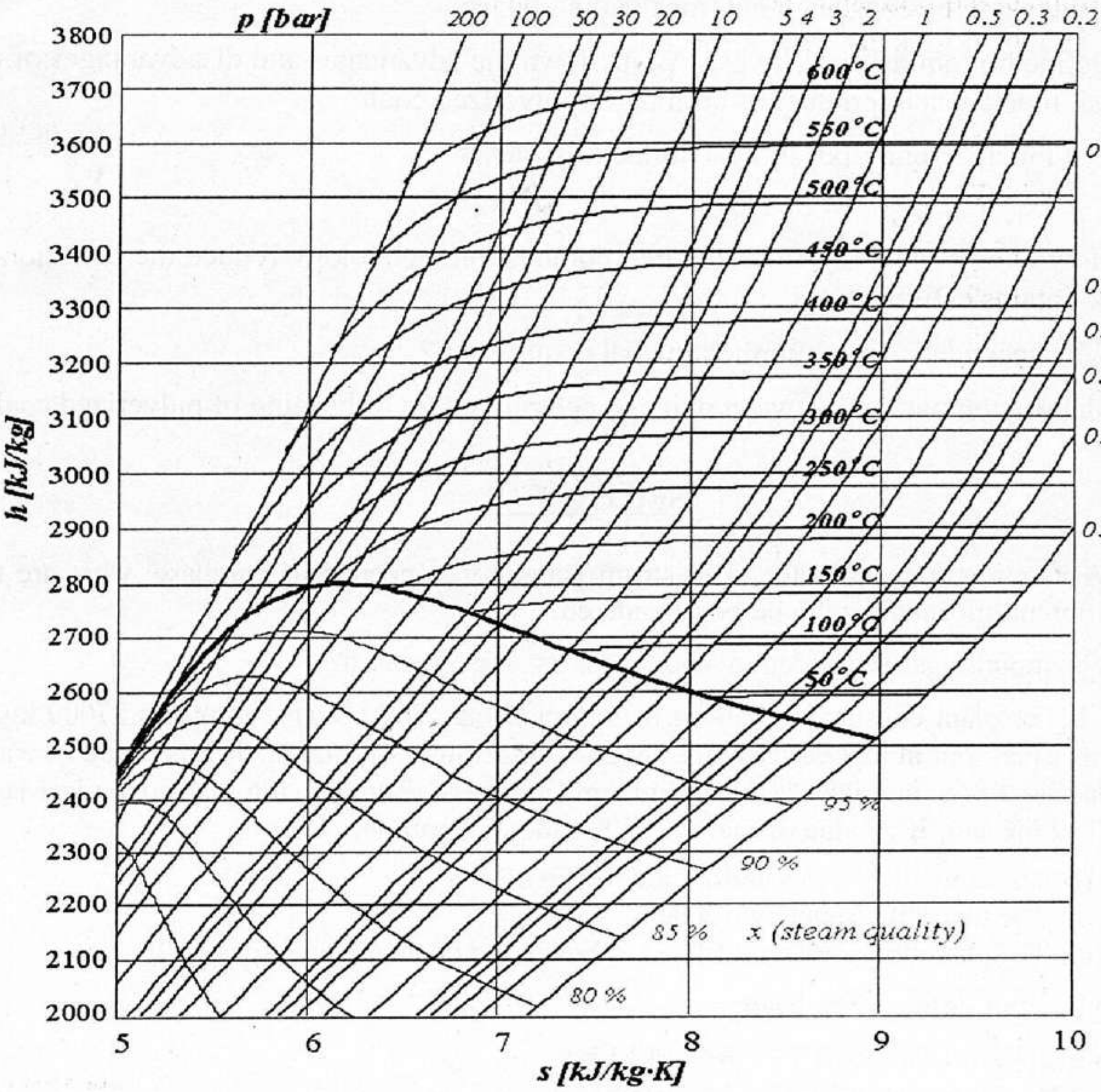
(i) Diagram efficiency (ii) stage efficiency (iii) Degree of reaction

5(c). In a four-stage steam turbine, the steam is supplied at 1st stage and passes through the intermediate stages and finally exhausted from turbine at 4th stage. The condition of steam and pressures are given below: 15

Stage	Condition	Pressure
1st stage supplied	300 °C	17.0 bar
2nd stage supplied	250 °C	7.5 bar
3rd stage supplied	170 °C	3.0 bar
4th stage supplied	0.97 dryness fraction	0.2 bar
4th stage exhaust	0.90 dryness fraction	0.1 bar

By using Mollier diagram, show the condition line for this multistage turbine and determine: (i) efficiency of each stage, (ii) overall efficiency, and (iii) reheat factor.

You may use the following Mollier Diagram.



6(a). How clean coal technologies are used to control the emission of pollutants? 08

6(b). Explain the Integrated Gasification Combined Cycle plant with CO_2 capture. 12

6(c). What does mean by CO_2 sequestration? Explain the 'Selective Non-catalytic Reduction' process to reduce the NOx emission from flue gas. 10

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